

MySQL Advanced

Ahmed Ehab Ahmed

Lab 1

1. Select the title, description, and length of all films that are longer than 120 minutes. Sort them from the longest to the shortest

```
SELECT title, description, length
FROM film
WHERE length > 120
ORDER BY length DESC;
```

2. Find all films that have a rental_rate of 0.99 or 2.99, but their replacement_cost is greater than 20.00

```
SELECT * FROM film
WHERE replacement_cost > 20 AND (rental_rate = 0.99 OR rental_rate = 2.99);
```

3. Count the total number of films available in each rating (G, PG, R, etc.)

```
SELECT rating, count(film_id) as count
FROM film
GROUP BY rating;
```

4. List the customer_ids who have made more than 30 separate payments in the payment table.

```
SELECT customer_id, count(customer_id) as count
FROM payment
GROUP BY customer_id
HAVING count(customer_id) > 30;
```

5. Get all "Cities" in the database and the "Country" they belong to, but only for cities located in 'Egypt'

```
SELECT city, "Egypt"
FROM city
WHERE country_id = (
    SELECT country_id
    FROM country
    WHERE country = "Egypt"
);
```

6. Display a list of all films and the names of the actors who starred in them. (show film id, title and actor name)

```
SELECT f.film_id, f.title, CONCAT(a.first_name, " ", a.last_name) as name
FROM film_actor fa
JOIN
    film f ON fa.film_id = f.film_id
JOIN
    actor a ON fa.actor_id = a.actor_id;
```

7. Find all customers who have rented a movie but haven't returned it yet. (show the customer name and the film title).

```
SELECT CONCAT(c.first_name, ' ', c.last_name) AS customer_name, f.title AS
film_title
FROM (
    SELECT customer_id, inventory_id
    FROM rental
    WHERE return_date IS NULL
) AS active_rentals
JOIN
    customer c ON active_rentals.customer_id = c.customer_id
JOIN
    inventory i ON active_rentals.inventory_id = i.inventory_id
JOIN
    film f ON i.film_id = f.film_id;
```

8. List the titles of all films whose length is greater than the average length of all films in the database.

```
SELECT title, length FROM film
WHERE length > (
    SELECT AVG(length) FROM film
);
```

9. Write a query to find the first_name, last_name, and email of customers who have zero rental records

```
SELECT first_name, last_name, email FROM customer
WHERE customer_id NOT IN (
    SELECT customer_id FROM rental
);
```

10. Create a view named `customer_spending_summary`. This view should display each customer's name, their total number of rentals, and the total amount of money they have paid.

```
CREATE VIEW customer_spending_summary AS
SELECT    CONCAT(c.first_name, " ", c.last_name) AS name,
          count(r.customer_id) AS total_number_of_rentals,
          sum(p.amount) AS total_amount_of_money
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id
JOIN payment p ON c.customer_id = p.customer_id
GROUP BY r.customer_id;
```

11. Use the previous view to find only customers who spent more than \$100

```
SELECT * FROM customer_spending_summary
WHERE total_amount_of_money > 100;
```

==> Built-in Function

1. Display actor names in the format: LAST_NAME, First_name (e.g., GUINNESS, Penelope).

```
SELECT CONCAT(
    UPPER(last_name),
    ', ',
    UPPER(LEFT(first_name, 1)),
    LOWER(SUBSTRING(first_name, 2))
) AS formatted_name
FROM actor;
```

2. Display all customer emails in lowercase and replace the domain `@sakilacustomer.org` with `@iti-students.edu`

```
SELECT LOWER(REPLACE(email, "@sakilacustomer.org", "@iti-students.edu"))
AS formatted_name
FROM customer;
```

3. Display the first 50 characters of each film's description followed by `"..."` and call the column `short_summary`.

```
SELECT CONCAT(SUBSTR(description, 1, 50), "...") AS short_summary
FROM film;
```

4. Find all customers who registered in the month of February (any year).

```
SELECT * FROM customer
WHERE MONTH(create_date) = 2;
```

==> User-Defined Functions

1. Create a function that takes actor_id and returns the concatenated first and last name of this actor.

```
DELIMITER /
CREATE FUNCTION get_actor_name(id INT)
RETURNS VARCHAR(100)
DETERMINISTIC READS SQL DATA
BEGIN
    DECLARE full_name VARCHAR(100);
    SELECT CONCAT(first_name, " ", last_name) INTO full_name
    FROM actor
    WHERE actor_id = id;
    RETURN full_name;
END /
DELIMITER ;
```

2. Create a function that takes customer_id and returns the total count of rentals made by this customer.

```
DELIMITER /
CREATE FUNCTION total_count_of_rentals(id INT)
RETURNS FLOAT
NOT DETERMINISTIC READS SQL DATA
BEGIN
    DECLARE count VARCHAR(100);
    SELECT COUNT(*) INTO count
    FROM rental
    WHERE customer_id = id;
    RETURN count;
END /
DELIMITER ;
```

3. Create a function that takes a DECIMAL value and returns it as a formatted string with a dollar sign (e.g., \$19.99).

```
DELIMITER /  
CREATE FUNCTION decimal_to_string(amount FLOAT)  
RETURNS VARCHAR(30)  
DETERMINISTIC NO SQL  
BEGIN  
    RETURN CONCAT("$", FORMAT(amount, 2));  
END /  
DELIMITER ;
```